FILM STUDIES
An Introduction

Ed Sikov

Second Edition

COLUMBIA UNIVERSITY PRESS NEW YORK

COLUMBIA UNIVERSITY PRESS
Publishers Since 1893
NEW YORK   CHICHESTER, WEST SUSSEX
cup.columbia.edu

E-ISBN 978-0-231-55156-4

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Library of Congress Cataloging-in-Publication Data
Names: Sikov, Ed, author.
Title: Film studies : an introduction / Ed Sikov.
Description: Second edition. | New York : Columbia University Press, [2020] | Series: Film and culture series | Includes index.
Identifiers: LCCN 2020001950 (print) | LCCN 2020001951 (ebook) | ISBN 9780231195928 (cloth) | ISBN 9780231195935 (paperback)
Subjects: LCSH: Motion pictures.
Classification: LCC PN1994 .S535 2020 (print) | LCC PN1994 (ebook) | DDC 791.43—dc23
LC record available at https://lccn.loc.gov/2020001950
LC ebook record available at https://lccn.loc.gov/2020001951

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Cover images: TCD / Prod.DB / Alamy Stock Photo

CHAPTER 13
FILM STUDIES IN THE ERA OF DIGITAL CINEMA

Putting together a second edition of this book has given me the chance to think about the ways in which film studies has changed over the past decade. When I wrote the first edition, I took the term *film studies* more or less literally: I assumed that the object of study was film—35 mm motion picture film to be exact. However, the advent of digital cinema has important implications for how we think about film studies. We now inhabit a world that Steven Shaviro and others have called “post-cinema,” a world in which film-as-film is no longer the dominant medium of our culture, having been replaced by digital and computer-based media. In many ways, everything has changed, and yet nothing has changed. The so-called digital revolution heralded a radical new era in film production, distribution, exhibition, and consumption. But underneath the surface of this revolution are many powerful currents linking the old and the new into a larger continuity. Chief among them is film form, the primary focus of this book.

WHAT IS DIGITAL CINEMATOGRAPHY?

Traditional cinematography and digital cinematography share an initial technological process: gradations of light enter a lens. In a process invented in the nineteenth century, used in old-fashioned still photography and, later, cinematography, the light hits a spool of plastic with chemicals on it—*film*—that exposes and records the patterns of light that have passed through the lens. When this film is developed using chemicals that react with the exposed film, a *negative* image is created; this negative is used, in turn, to create a *positive* image.

In digital cinematography, there is no spool of plastic—no film. The light hits a device called an image sensor that converts the light into tiny *pixels*. Each pixel corresponds in brightness and color to its place within the visual field captured by the lens. The visual information of each pixel is stored and then reassembled as numerical data (the digits that give digital cinematography its name). The final product of a digital image is not an impression of light on film, as it is in traditional cinematography, but rather a kind of mosaic of tiny bits of data that are converted into pixels and reassembled into images that are projected onto a screen.

THE IMPACT OF DIGITAL TECHNOLOGY ON FILMMAKING

In many ways, the shift from shooting feature films on 35mm celluloid film to shooting them with digital cameras is nothing short of revolutionary. In material terms, directors no longer have to worry about such nuisances as dust building up in the camera—dust that could ruin an otherwise perfect take by scratching the film or simply by getting between the lens and the film and blocking some of the light from registering on the film. Distributors, too, find great benefits in digital cinema. They no longer have to ship cumbersome cans of 35mm film to theaters, where the risk of scratches is even greater. Furthermore, when a 35mm print is run repeatedly through a film projector, it necessarily deteriorates over time. Now, theaters can screen digitized films hundreds of times without having to worry about the prints' condition, *because they are no longer prints*.

In 2013, most of the major film distributors announced flatly that they would no longer distribute 35mm film. This decision was also revolutionary. Why? Because if films are no longer on film, what should we call them? They could still legitimately be known as *movies* because they are still motion pictures, but since the material has shifted from plastic film to digits, the very nature of the art form has fundamentally transformed. To many people in the filmmaking community, this change has both positive and negative consequences.

From the perspective of film analysis, there is little or no difference between digital cinema and 35mm motion picture film; film language remains the same. The sequence of shots in a film remains the same whether it is shot on film or digitally. The same is true for a film seen on a large screen in a movie theater and for a film seen at home on television, a computer, a tablet, or even on a cellphone. If we're watching *Casablanca*, it will be exactly the same, shot for shot, whether we watch it on a big or a small screen, in a theater or on the bus. In this sense, digital cinema does not transform film language. The basic formal techniques associated with mise-en-scene, editing, and sound remain the same.

However, digital tools can extend or enhance certain aspects of formal technique. Long takes, for example, can be longer when shot digitally since they are not limited by the size of the film magazine. Digital technology also enables special effects to be more photorealistic in nature, especially in the area of IMAGE COMPOSITING. At the same time, digital filmmaking technology has made the means of production of films available, at relatively minimal cost, to a greater number of would-be filmmakers than the relatively expensive, 35mm filmmaking technology of the past, though it still takes a considerable amount of money to market a film to the general public.

EXPANDING THE FILMMAKING COMMUNITY

There is no doubt that shifting away from bulky 35mm celluloid cameras to digital cameras greatly expanded the number and variety of people who were able to make movies. From the years when you had to have enough money to rent a 35mm camera before you could even begin to make a movie, suddenly it became possible to make one using only a digital camera.

In fact, a number of very good films have been made using iPhones. These films include (among others) *Night Fishing* (2011), shot by the Korean directors Chan-kyong Park and Chan-wook Park using an iPhone 4; *Tangerine* (2015), shot by Sean Baker using three iPhone 5S devices; and *Unsane* (2018), shot by Steven Soderbergh using an iPhone 7 Plus. Now that anyone with a smartphone can make a movie, the variety of filmmakers and the subjects of their films have greatly expanded. *Tangerine* may not be the first movie about transgender women, but it might very well have not gotten the financing needed to make it if it hadn't been shot relatively cheaply with an iPhone. Telling the story of Sin-Dee, a *transgender* prostitute who discovers that her pimp has been cheating on her with a *cisgender* woman, *Tangerine* is a comedy-drama with a cast of unknowns; it's not the kind of movie that big studios tend to GREENLIGHT. It is a genuinely INDEPENDENT FILM, meaning that it was financed independently, not merely that it seems to be cutting-edge.

DIGITAL CINEMA AND FILM THEORY

The French film theorist André Bazin once wrote that the cinema, in addition to being an ontology, was also a language. Ontology is the philosophical study of being. What Bazin meant by ontology is a complicated subject, but his distinction between ontology and language is a very useful way of thinking about the two major axes that comprise the study of the cinema. One axis is SYNCHRONIC; it lays out a series of relationships at a specific moment between the spectator and the image. This axis extends from the image back to what it is an image of. For example, you see a tree onscreen, and you know—having seen many trees in your life—that is in fact the image of a tree. The other axis is DIACHRONIC and involves a much simpler series of relationships—those which occur over time from shot to shot. Film language is situated along this axis; it is here that the spectator does the work of mentally assembling the succession of shots in a scene, sequence, or entire film into a meaningful whole.

Although the nature of film language translates easily from analog to digital cinema on the diachronic axis, the same is not the case for Bazin's ontological axis—the one that connects us to whatever stood before the camera when the image was created. The relationships that constitute this axis are dramatically different when it comes to comparing analog and digital cinema. According to Bazin, whatever is placed before the camera deposits traces of itself (in the form of light reflected off of it) onto the camera negative. For Bazin, “the photograph as such and the object in itself share a common being, after the fashion of a fingerprint.” The language Bazin uses is that of a transfer: “Photography transfers reality from the object depicted to its reproduction.” In other words, it’s not only light that transfers from the object being filmed; it is reality itself. It is a transfer made possible by the movement of light from the filmed object to its reproduction on a photographic plate. For Bazin, it is that light that testifies to the existence of that object—that extends the existence of that object from one point to another, that proves its reality. In Bazin’s fantastic formulation, the world’s reality proves itself by being photographed.

With digital imaging, Bazin’s “transfer” of light gives way to a “translation” of light. The image sensor in a digital camera translates light waves into electrical signals and then samples those signals. At the same time, the luminance values—the intensity of light—of that signal are measured and quantized in terms of their amplitude. In short, the image sensor transforms the continuous data within a stream of light into a series of discrete numbers. It is in this translation of image into numbers that digital imaging’s greatest strength and most serious shortcomings lie. Its underlying foundation in numbers makes the digital image both stable and unstable. In the form of numerical data, digital images are extremely stable; they can be copied without degradation and readily transported over the internet. But those images can also be quite easily changed by altering the numbers that constitute them. As the director Wim Wenders put it, “The digitized picture has broken the relationship between picture and reality once and for all. We are entering an era when no one will be able to say whether a picture is true or false.”

POST-TRUTH?

Some call this the era of Photoshop; others, more provocatively, call it the "post-truth era." Even before Photoshop, in 1982, the *National Geographic* altered the positions of the pyramids at Giza to obtain a more pleasing composition for the cover of its magazine. What some might see as a deliberate falsification of history, others would defend as a simple artistic choice. In the last week of June, in 1994, both *Newsweek* and *Time* magazines published the *same* police mug shot of O. J. Simpson on their covers. While *Newsweek* ran the photo more or less as it was, *Time* extensively retouched it, darkening Simpson's skin tone to make him blacker and, therefore, according to many observers, more sinister; by almost any account, the image's racism increased to a profound degree.

Though analog photography is also subject to manipulation (and consequent falsification), it has also functioned as legal and scientific evidence for over 150 years. In *The Reconfigured Eye: Visual Truth in the Post-Photographic Era* (1994), William J. Mitchell argues that an analog "photograph passes for incontrovertible proof that a given thing happened." But he goes on to note that the traditional truth claims of analog photography are "subverted by the emergence of digital imaging."

For its part, digital photography lacks the claims of truth that analog photography may make. In recent years, this has become increasingly apparent on the internet, which has become a platform for disinformation and fake news. Cyberspace is a minefield of a unique form of digital cinema known as "deep fakes." "Deep fakes" draw on machine learning to analyze existing videos of a person and combine them into a new, synthetic video that depicts the subject doing or saying things that never happened in reality. In April 2018, Jordan Peele, the writer and director of *Get Out* (2017), and his brother-in-law Jonah Peretti, BuzzFeed's CEO, put together a "deep fake" that went viral on BuzzFeed. Presented as a public service announcement about the danger of deep fakes, their film features former president Barack Obama seemingly warning viewers that "we're entering an era in which our enemies can make it look like anyone is saying anything at any point in time, even if they would never say those things," such as "Ben Carson is in the sunken place!" and "Donald Trump is a total and complete dipshit." At the end of the one-minute clip, Obama—who in reality has been voiced by Peele—warns the audience to "stay woke" (a quote from *Get Out*) when watching videos online.

Even though the bonds between digital cinema and reality are often tenuous, digital cinema can look quite real. That is in large part because it was designed to look like film. In duplicating the "look" of 35mm film, digital cameras did not so much strive for realism as for photorealism—the look of photographed reality. For example, the sensor arrays at the heart of digital imaging technology were developed in response to cinematographers' requests for an image with a DEPTH OF FIELD that better resembles that of film cameras. The computer chips initially used in digital cameras of the mid- to late 1990s were quite small in size—roughly one- to two-thirds of an inch in size. As a result, the images they generated had tremendous depth of field. In fact, the depth of field was so great that it looked unnatural to audiences accustomed to analog cinematography. Cinematographers complained and, as a result, computer chips (or charge-coupled devices) have grown in size, from 8 and 16 millimeters—roughly 40 percent the size of 35mm film—to a full 35 millimeters in order to provide the same angle of view and depth of field as 35mm cinematography. At the same time, in response to the concerns of the American Society of Cinematographers, the designers of digital cinema cameras equipped them with "set-up cards" that enabled the cameras to duplicate the optical characteristics of the most popular Kodak film stocks. Even though digital imaging permitted a manipulation of the image that was potentially radical in nature, the great majority of digital films did not pursue the revolutionary potential of the new medium, seeking instead to simulate the photorealism of traditional motion picture cinematography.

In the first issue of *Wired* (1993), its editors embraced the notion of a digital revolution, which they described as "whipping through our lives like a Bengali typhoon." Paraphrasing the Canadian media theorist Marshall McLuhan, who coined the phrase "the medium is the message" and whom they referred to as their "patron saint," the editors of *Wired* declared: "The medium, or process, of our time—electric [that is, digital] technology is reshaping and restructuring patterns of social interdependence and every aspect of our personal life. It is forcing us to reconsider and re-evaluate practically every thought, every action, and every institution formerly taken for granted. Everything is changing: you, your family, your education, your neighborhood, your job, your government, your relation to others. And they're changing dramatically." Some people went so far as to dub the digital revolution the Third Industrial Revolution, likening it to the sweeping changes brought about by the shift from handmade to machine-made goods that took place in the First Industrial Revolution and the rapid industrialization that characterized the Second. The shift from analog to digital technology did, in fact, transform the ways in which we interact with one another and the world. Our reliance on computers, the internet, cell phones, and smartphones provides ample evidence of that. In 2018, the average American spent 24 hours a week on the internet (up from 9 hours in 2000); from 2010 to 2018, the proportion of people accessing the internet through their smartphones rose from 23 to 84 percent; streaming was responsible for $2.5 billion worth of music sales in the United States in 2017; and online video accounted for nearly 80 percent of all web traffic in 2018.

But the digital revolution was also hyped in ways that overstated its transformative nature. The so-called digital revolution that took place in the motion picture industry provides a good example of such hype. Hollywood exploited the notion of digital technology as a radical break from earlier analog filmmaking in an effort to market the new technology to its dwindling audience base, using it as a novelty to lure audiences back into theaters. In the 1990s, moviegoers, having just ridden the wave of the transition in the music industry from vinyl (analog) to CD (digital), embraced the digital turn in the cinema with enthusiastic anticipation. The word "digital" on a movie theater marquee resonated with the core of moviegoers whose ages ranged from twelve to twenty-four and who accounted for 39 percent of all tickets sold. This age group lived in a digital world, and the movies had finally caught up with the digital revolution that had already taken over other major forms of mass media. It was to this digital generation of the nineties that Hollywood pitched digital cinema as a revolutionary step forward in the history of the motion picture industry, likening it to earlier technological revolutions such as the coming of sound, color, and widescreen. At the theatrical opening of George Lucas's all-digital *The Phantom Menace* in 1999, lobby displays listed the major "milestones in motion picture history," from the invention of motion pictures through the transition to sound, color, and widescreen, and culminating with the premiere of digital cinema. One of the producers of *The Phantom Menace*, Rick McCallum, boasted that "like the introduction of sound and color, these digital screenings represent the beginning of a new era in film presentation."

Russell Wintner of CineComm Digital Cinema likened *The Phantom Menace* to *The Jazz Singer*, the 1927 Warner Bros. film that featured both a synchronized recorded musical score and lip-synchronous singing and speech in certain sequences. To this digital cinema executive, the arrival of digital was like the coming of sound.

In fact, digital cinema was not revolutionary in the way that those other technological revolutions were. Digital projection did not in any way transform the nature of the motion picture experience. It was not like the transition from silent to sound cinema which introduced a new dimension to the moviegoing experience. Nor did it resemble Cinerama, for example, which produced a dramatic sense of audience participation amongst spectators who, surrounded by image and sound, enjoyed the illusion of having entered the space of the picture. Audiences viewing digital projection did not experience the cinema differently. In fact, digital projection was designed to simulate the projection of 35mm film but to do so without its occasional flaws like emulsion scratches, dirt, or JITTER. Digital technology did provide filmmakers with an unprecedented means of manipulating the moving image during production and postproduction, but audiences watching digital cinema in the theater were not as blown away by the experience as the promotional hype suggested they would be. What they were watching was still primarily conventional cinema.

Digital cinema was certainly not revolutionary if one measures a revolution by the speed at which it occurs. The transition to sound, for example, took only two or three years (ca. 1926–1929). The conversion of the film industry to widescreen was also relatively rapid (ca. 1952–1954). But the shift from analog to digital cinema took more than thirty-five years (from around 1977 until 2012) in large part because, unlike other technological "revolutions" that involved only a single technology, digital cinema was a multi-technological phenomenon. It came into being piece by piece as its technology was adopted in various sectors of the motion picture industry and across the full range of motion picture production, distribution, and exhibition. The term *digital cinema* properly encompasses the digitization of each aspect of the film chain, from production and postproduction (editing) to distribution and exhibition (projection). Digital cinema cannot be said to have been fully adopted by the film industry until digital technology had taken over the entire chain of film production, distribution, and exhibition.

So it was not until *The Phantom Menace* (1999) that films began to be exhibited digitally. And it was not until *Avatar* (2009) that a majority of screens around the world had converted from analog to digital projection. Finally, it was not until 2012 that the film industry ceased making films available to theaters on 35mm film.

Digital technology was employed in the late 1970s with the development of computer-controlled cameras for filming special-effects sequences in films such as *Star Wars* (1977). In 1982, Industrial Light & Magic developed technology for creating images on a computer for the Genesis sequence in *Star Trek II: The Wrath of Khan*. In 1984, George Lucas and his team of engineers developed EditDroid and SoundDroid—computerized, electronic, nonlinear editing systems. In the 1980s, James Cameron relied increasingly on computer-generated imagery (CGI) for special-effects work in *The Terminator* (1984) and *The Abyss* (1989), resulting in the three-dimensional "pseudopod" character in *The Abyss* and the "liquid-metal man" in *Terminator 2* (1991). The early 1990s also witnessed an industry-wide shift in postproduction from LINEAR to NONLINEAR EDITING. At the same time, Hollywood introduced

5.1 digital sound, with left, center, and right speakers behind the screen, left and right surround speakers at the rear of the auditorium, and a subwoofer for low frequency sound effects at the front of the auditorium. In 1995, Walt Disney Pictures released *Toy Story*, the first completely computer-generated film. By the end of the decade, Lucas and others had begun filming live-action scenes with digital cameras. *The Phantom Menace* (1999) contained over two thousand digital effects shots, and *Attack of the Clones* (2002) was shot entirely on 24-frame, progressive high-definition digital video. The completion of the so-called digital revolution in the cinema occurred in June 1999 when *The Phantom Menace* was distributed and exhibited in electronic form. Digital projection marked the completion of the film chain, but it also exposed the weakest link in the chain. Exhibitors balked at the cost of digital projectors, which were, at that time, $100,000 per unit compared with $35,000 for a film projector. The exhibition phase brought the digital revolution to a grinding halt because it lacked what we now refer to as a “killer app.”

While digital cinema’s chief advocates touted its novelty value, digital cinema technology was busy simulating older analog technology. By 2004, it had become obvious to key industry advocates of digital technology (such as James Cameron, George Lucas, and Jeffrey Katzenberg, the CEO of DreamWorks Animation) that the digital revolution had stalled in part because it had little or no novelty value. In an attempt to compensate for this problem, proponents of digital cinema tried to give digital cinema a novelty phase through the hybrid technology of digital 3D. This served as digital cinema’s killer app. Digital 3D marks an attempt on the part of the film industry to artificially manufacture a novelty phase for digital cinema. But, although it does give digital cinema a novelty value—something it *could* do that conventional cinema could *not*—it is not a genuine novelty. 3D is not new; its technology has been around since the invention of stereoscopic photography in 1838 and was pioneered in motion pictures by William Friese-Greene in the 1890s. But advocates of digital cinema tried digital 3D, and it worked. The stalled rollout of digital cinema gave way to dramatic expansion in the number of digital screens, both domestic and international. For various reasons, among them the 3D digital craze that followed the phenomenal success of *Avatar* (2009), which earned over $2.7 billion worldwide, and 2010's *Alice in Wonderland* and *Toy Story 3*, which earned over $1 billion each, the conversion of theaters to digital has increased dramatically, doubling each year since 2006. In 2006, there were only 2,991 digital screens worldwide; by 2017, there were 619,275.

GOODBYE MULTIPLEXES?

Digital cinema has succeeded as a new technology by borrowing its identity from other, analog technologies—from 35mm film, which it has sought to emulate in terms of its *look*, and from 3D stereophotography, whose novelty value it essentially borrowed in an attempt to become a new norm. The history of technology in the cinema is a constantly shifting series of identities it has taken up from the different technologies that have informed its development.

Digital cinema has ultimately proven itself to be revolutionary, but not in the ways that its proponents had originally envisioned. If it has transformed the cinema-going experience, it has done so by providing an alternate experience that has slowly taken old-fashioned cinema's place. With the proliferation of digital platforms capable of delivering the cinema to consumers when and where they want, the cinema is no longer tied to the movie theater with its enormous screens that display life-size (or bigger) images to a mass audience. The digital revolution is a revolution not in *what* the cinema is but in *where* the cinema takes place.

Of course, television has provided an alternate space for experiencing cinema ever since it was mass-marketed to consumers in the home in the late 1940s. But digital technology enables the cinema to take place not just on a fixed screen in the home but on a variety of mobile screens that could be taken virtually anywhere. This has been made possible by the advent of STREAMING. The term *streaming* refers to multimedia that is delivered by a provider by way of the internet to an end user (or to put it quaintly, a *person*). The term refers to the delivery method of the medium rather than the medium itself: movies can be streamed, live sports events can be streamed, music concerts can be streamed, and so on. Streaming serves as an alternative to *file downloading*, whereby the consumer must receive the whole file before watching or listening to it. The streaming of video has, in turn, been made possible by the development of extremely powerful and efficient compression ALGORITHMS, which squeeze the raw data originally captured by the image sensor into manageable files that can be streamed over the internet. Each frame of a 4K digital cinema image contains 310,728,960 bits, or nearly 40 megabytes of data; a two-hour movie (172,800 frames at 24 frames per second) would amount to 7 terabytes (or 1,024 gigabytes) of data. A standard dual-layer DVD can store only 9.4 gigabytes. It would therefore take over seven hundred DVDs to store a two-hour movie in its uncompressed form, and it would take over two weeks to transmit such a film by satellite to a digital server in a movie theater. Compression is obviously required to reduce the total amount of data present in the original digital image recorded by the camera sensor.

As a medium that has been reengineered so that it can be transported over the internet, the cinema has become one piece of a much larger media infrastructure designed for the global transportation of data—an infrastructure known as the “information superhighway” twenty years ago but has since fallen into disuse. Globally, video accounts for 57.7 percent of all traffic on the internet, web browsing for 17 percent, gaming 7.8 percent, and social media 5.1 percent. Clearly, not all of this traffic in video is cinema. Netflix does consume 15 percent of all global bandwidth, but the bulk of what its viewers watch is TV shows, which account for 70 percent of Netflix’s business. As the film historian Stephen Prince points out, as the consumption of the cinema shifts from movie theaters and living rooms, where it plays on nonportable, relatively heavy equipment like digital cinema projectors and DVD and Blu-ray players, to tablets and smartphones, the existence of a material object that one might identify as a movie has become increasingly difficult to establish. DVDs and Blu-ray discs are disappearing; indeed, the average computer no longer has an internal DVD disc drive, an accessory that was absolutely necessary just a few years ago.

In the age of celluloid, one could define a motion picture as a physical object consisting of images and sounds that have been fixed on a celluloid base. In the digital era, a motion picture consists of a stream of data files that are only assembled at the time and place of their consumption. Streaming has dematerialized the cinema.

For many film historians and theorists, the cinema was born when life-size images were projected on a large movie screen for a mass audience. They defined cinema as an APPARATUS (see the discussion of the apparatus in the "Film and Ideology" chapter). That apparatus continues to define much of digital cinema, which relies on theatrical exhibition for 25 percent of its revenues. But 75 percent of the audience for any given film sees it outside of a theatrical context—on screens with diminished image sizes and with mostly monaural (rather than stereo 5.1 surround) sound. If the film is displayed in its original aspect ratio (see chapter 3), viewers will see the same film, shot for shot, that was originally screened in the theater, but they will not have the same experience of the image and its accompanying sounds as a theatrical moviegoer has. An image that was anywhere from thirty to sixty feet wide in a movie theater shrinks in size on a smart phone to four to six inches (measured diagonally). In the theater, a viewer can be absorbed by the movie into its onscreen world; with a smartphone, watching a movie is more like peeping into another room from the outside through a tiny keyhole.

Moreover, the activity of watching a movie is no longer communal. Unless you are seeing a real stinker, you are probably not alone in a conventional movie theater. Usually there are people all around you, and their emotional responses to the images they see projected on the screen and the sounds that are blasted through the speakers—from raucous laughter to sniffling to gasps of terror—are key to forming your own. How different that experience is from watching a movie by yourself on a smartphone. Isolated from the rest of the world, experiencing it all by yourself, you aren't getting the emotional cues afforded by a surrounding audience. This isolation can prove to be very damaging. Sure, you get to watch the movie anytime and anywhere you want, but you are experiencing it in a greatly shrunken form, all alone. And you can start it and stop it randomly to take a phone call or write a text message, so there is no continuity to your experience.

The director David Lynch (*Mulholland Dr.*, *Blue Velvet*) has gone so far as to declare, “If you are playing the movie on a telephone, you will never in a trillion years experience the film. You’ll think you’ve experienced it, but you’ll be cheated. It’s such a sadness that you think you’ve seen a film on your … f *****g telephone. Get real.” The director Robert Zemekis (*Back to the Future*) suggests that watching movies on a cell phone threatens to change how an audience sees a film: “I don’t think they appreciate the attention to detail. Maybe subconsciously they feel it, but maybe they don’t. Having a perfectly composed shot doesn’t matter if you are watching it on an iPhone, does it? You wouldn’t see it.”

In addition to some filmmakers, a number of film archivists also look skeptically upon digital imaging technology and its promise of eternal life. In its 2007 report on issues related to the archival preservation of digital cinema, *The Digital Dilemma*, the Academy of Motion Picture Arts and Sciences concluded that there was “no digital archival master format or process with longevity characteristics equivalent to that of film” and that there was no “operationally and economically sustainable means to maintain long-term access to [digital] materials.” Paolo Cherchi Usai, the former senior curator at the George Eastman Museum’s moving image department, warns that “digital technology offers the seductive promise of a real miracle: perfect vision, eternal moving images that can be reproduced ad infinitum with no loss of visual information—as blatant a lie as the claim that compact discs and CD-ROMs will last a lifetime.” If hope for the future of the cinema lies in the hands of digital technology, we need to recognize not only its many virtues but also its considerable limitations.